

Decoding the Cloud Security Alphabet Soup

Charlie Winckless

© 2026 Gartner, Inc. and/or its affiliates. All rights reserved. Gartner is a registered trademark of Gartner, Inc. and its affiliates. This publication may not be reproduced or distributed in any form without Gartner's prior written permission. It consists of the opinions of Gartner's research organization, which should not be construed as statements of fact. While the information contained in this publication has been obtained from sources believed to be reliable, Gartner disclaims all warranties as to the accuracy, completeness or adequacy of such information. Although Gartner research may address legal and financial issues, Gartner does not provide legal or investment advice and its research should not be construed or used as such. Your access and use of this publication are governed by [Gartner's Usage Policy](#). Gartner prides itself on its reputation for independence and objectivity. Its research is produced independently by its research organization without input or influence from any third party. For further information, see ["Guiding Principles on Independence and Objectivity."](#)

Gartner®

Is It Alphabet Soup?



... Or Acronym Soup?



Cloud Security Elements

Covered

- Workload security
- Identity and access security
- Platform and Configuration control
- DevSecOps and code security
- Resilience

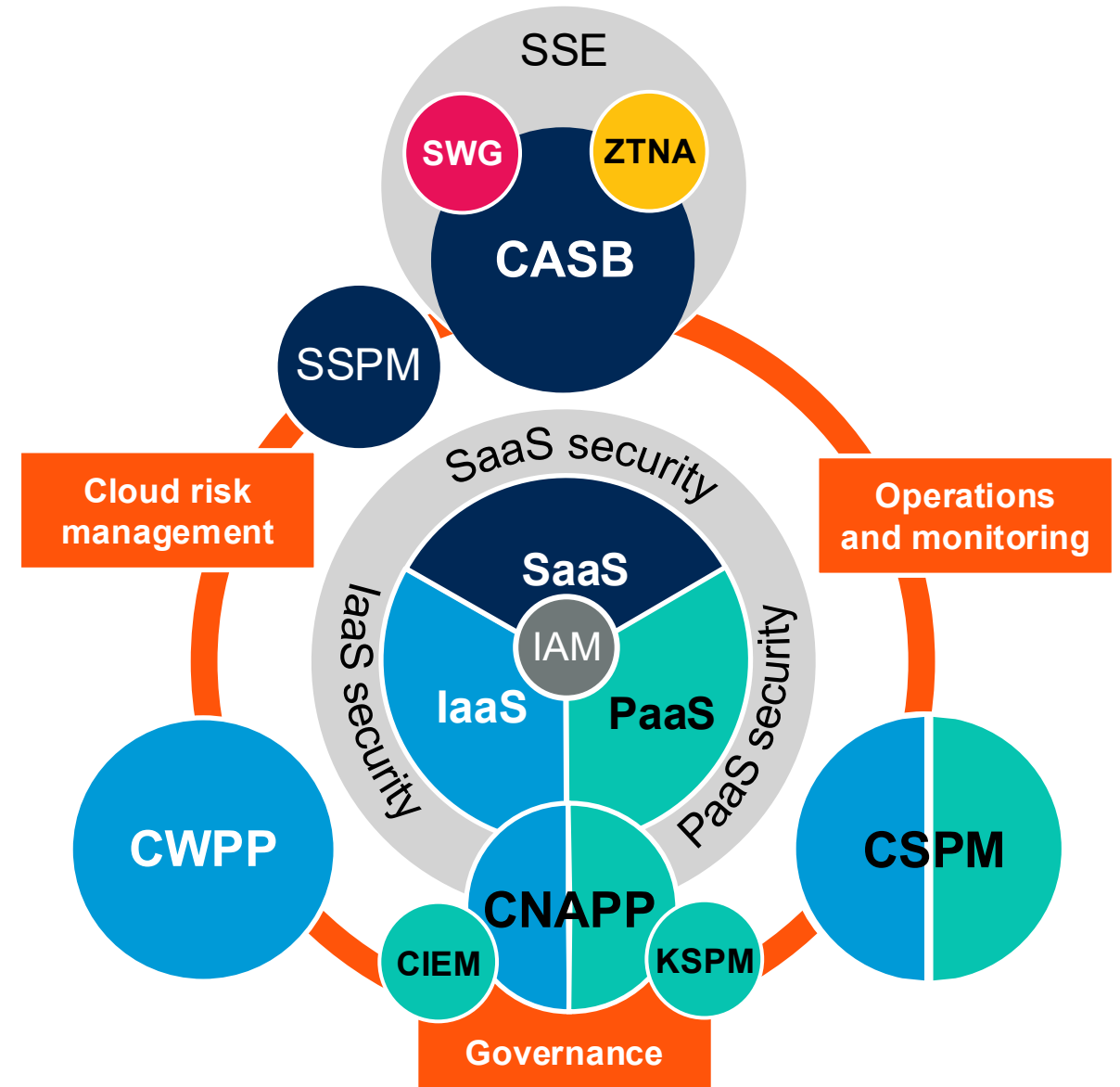
Not covered

- Firewalls and network ACLs
- Web security controls
- Load balancers
- Patching



**Network security controls
are **a very small part**
of cloud security.**

- **CSP** — cloud service providers
- **CASB** — cloud access security broker
- **CIEM** — cloud identity and entitlement management
- **CWPP** — cloud workload protection platform
- **CSPM** — cloud security posture management
- **CNAPP** — cloud-native application protection platform = CWPP + CSPM + CIEM + IaC
- **IaaS** — infrastructure as a service
- **IAM** — identity and access management
- **KSPM** — Kubernetes security posture management
- **PaaS** — platform as a service
- **SaaS** — software as a service
- **SSPM** — SaaS security posture management
- **SSE** — secure service edge = CASB + SWG + ZTNA
- **SWG** — secure web gateway
- **ZTNA** — zero-trust network access
- **FaaS** — function as a service





Shared Responsibility

How do I know who is **responsible
for securing what's in the cloud?**

Shared Responsibility Model

Customer responsibility
 Shared or contingent on deployment pattern
 Cloud provider responsibility

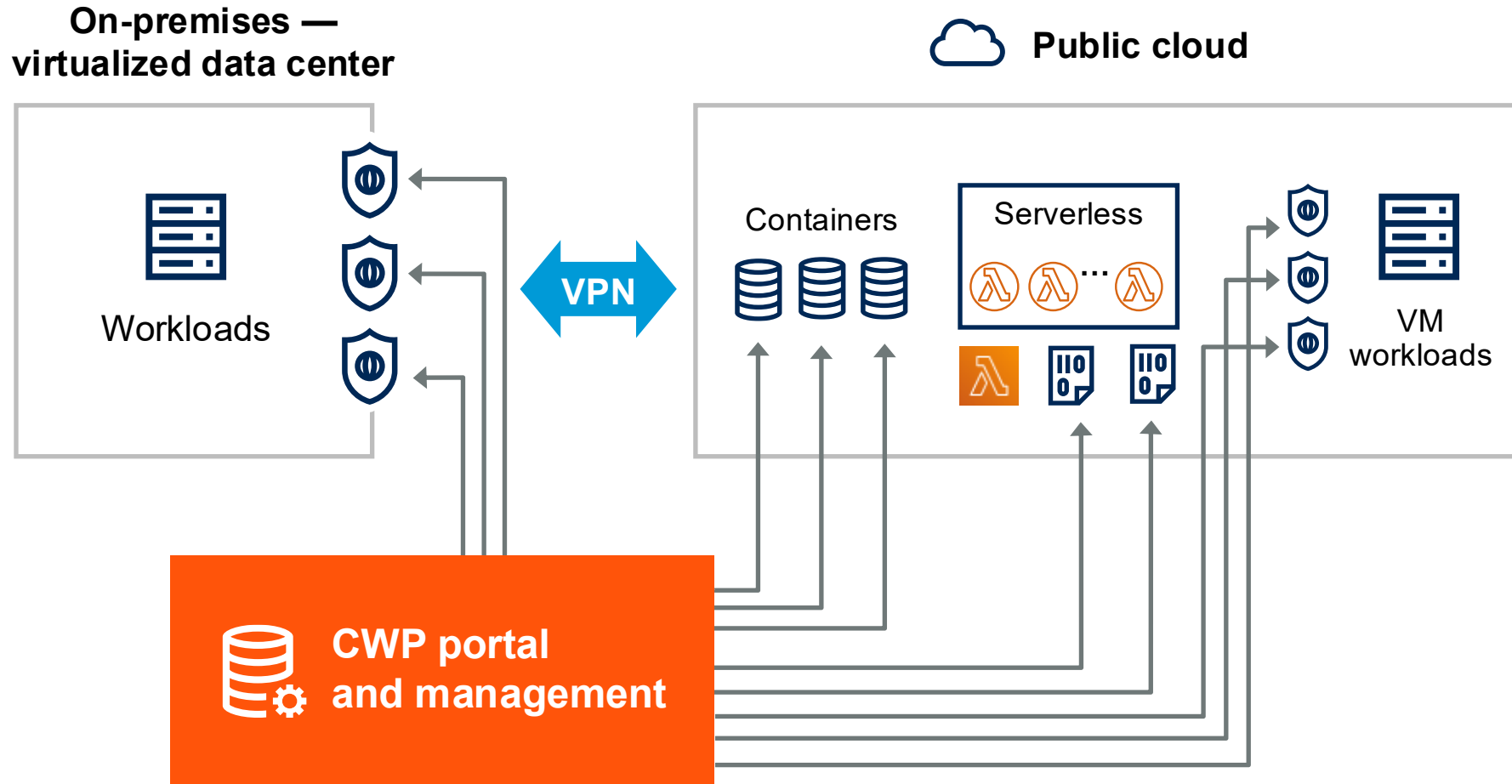
	Private/on-premises	IaaS	CaaS	FaaS	PaaS	SaaS
Business continuity	Customer responsibility	Customer responsibility	Customer responsibility	Customer responsibility	Customer responsibility	Customer responsibility
Identity and access management	Customer responsibility	Customer responsibility	Customer responsibility	Customer responsibility	Customer responsibility	Customer responsibility
Data	Customer responsibility	Customer responsibility	Customer responsibility	Customer responsibility	Customer responsibility	Customer responsibility
Application	Customer responsibility	Customer responsibility	Customer responsibility	Customer responsibility	Customer responsibility	Shared or contingent on deployment pattern
Application API	Customer responsibility	Customer responsibility	Customer responsibility	Customer responsibility	Customer responsibility	Shared or contingent on deployment pattern
Workload	Customer responsibility	Customer responsibility	Customer responsibility	Shared or contingent on deployment pattern	Shared or contingent on deployment pattern	Cloud provider responsibility
Virtual network	Customer responsibility	Customer responsibility	Customer responsibility	Cloud provider responsibility	Cloud provider responsibility	Cloud provider responsibility
Service orchestration	Customer responsibility	Customer responsibility	Shared or contingent on deployment pattern	Cloud provider responsibility	Cloud provider responsibility	Cloud provider responsibility
Virtualization/cloud infrastructure	Customer responsibility	Cloud provider responsibility	Cloud provider responsibility	Cloud provider responsibility	Cloud provider responsibility	Cloud provider responsibility
Physical	Customer responsibility	Cloud provider responsibility	Cloud provider responsibility	Cloud provider responsibility	Cloud provider responsibility	Cloud provider responsibility



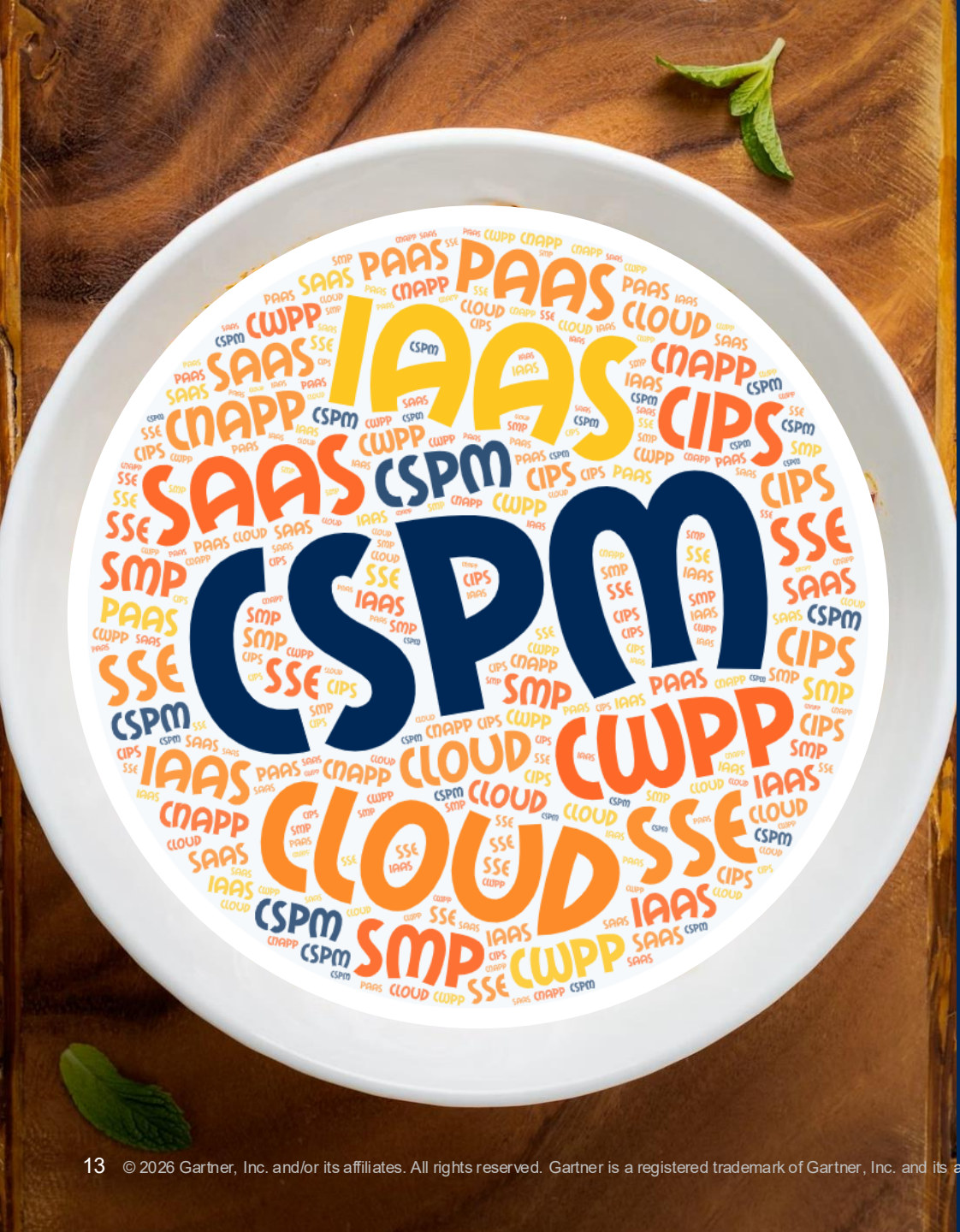
Workload Controls

**Diverse and dynamic workloads need
visibility and **some** protection at
runtime.**

Cloud Workload Protection



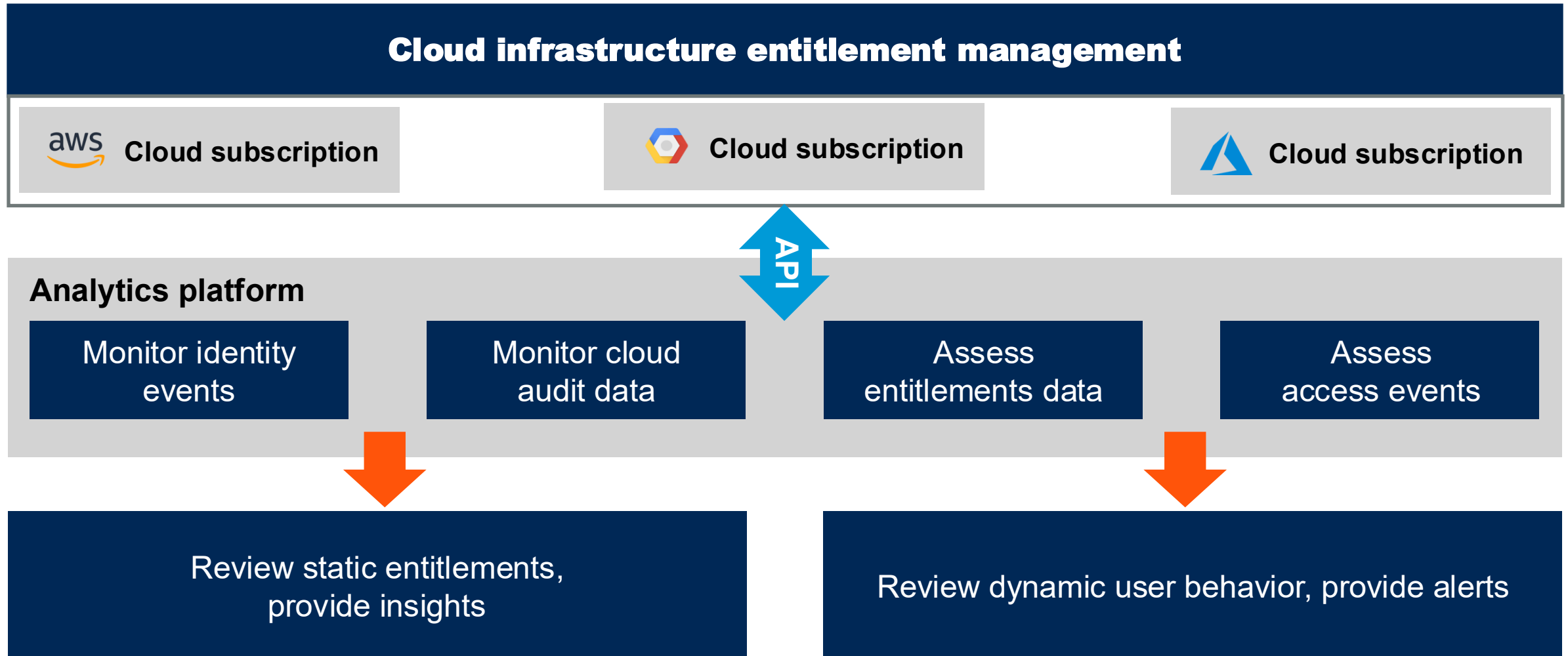
- VMs have agents deployed
- Serverless can use:
 - Embedded agents/enforcement layer
- Containers can use:
 - K8s host agents
 - Sidecars
 - DaemonSets
 - Kernel agents
 - eBPF



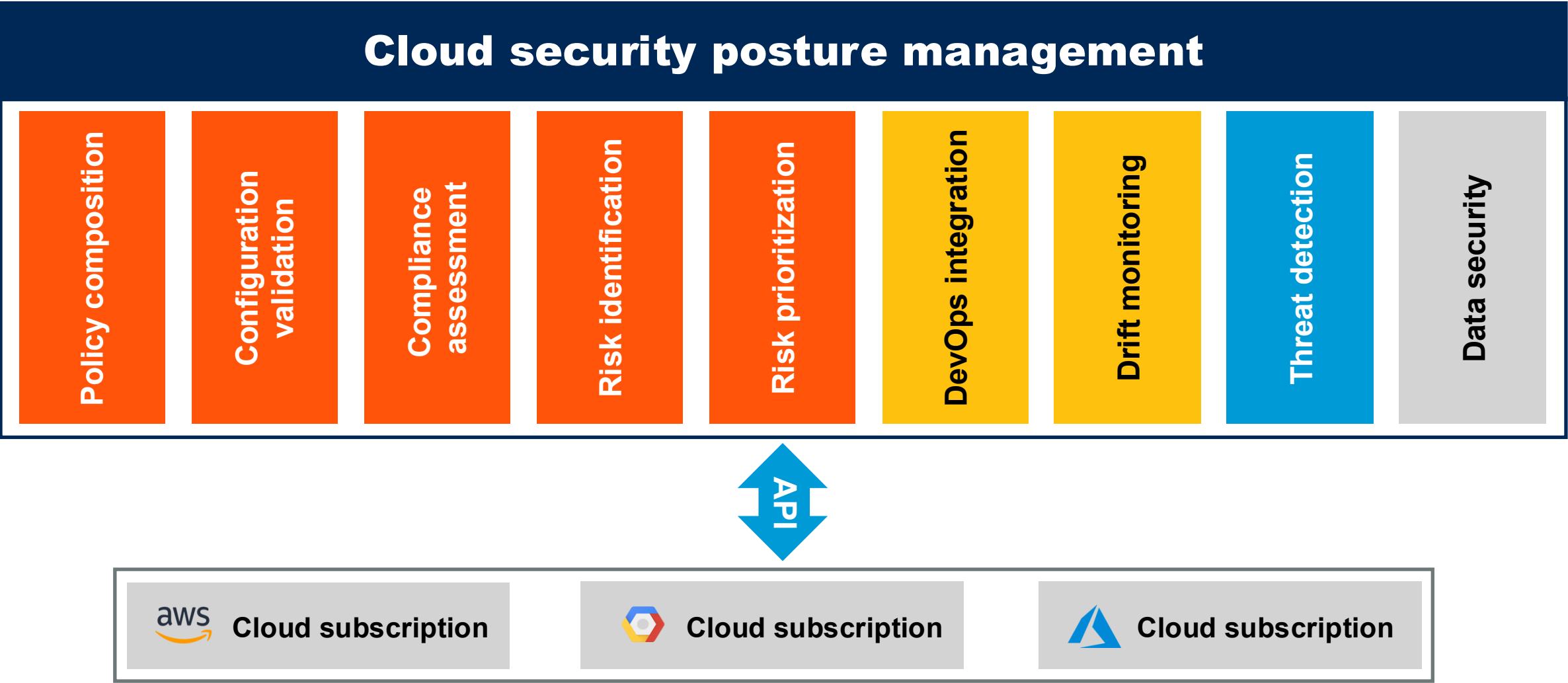
Platform Controls

**Cloud configurations are complex,
dynamic and are the **primary**
attack surface; misconfigurations
exacerbate impact.**

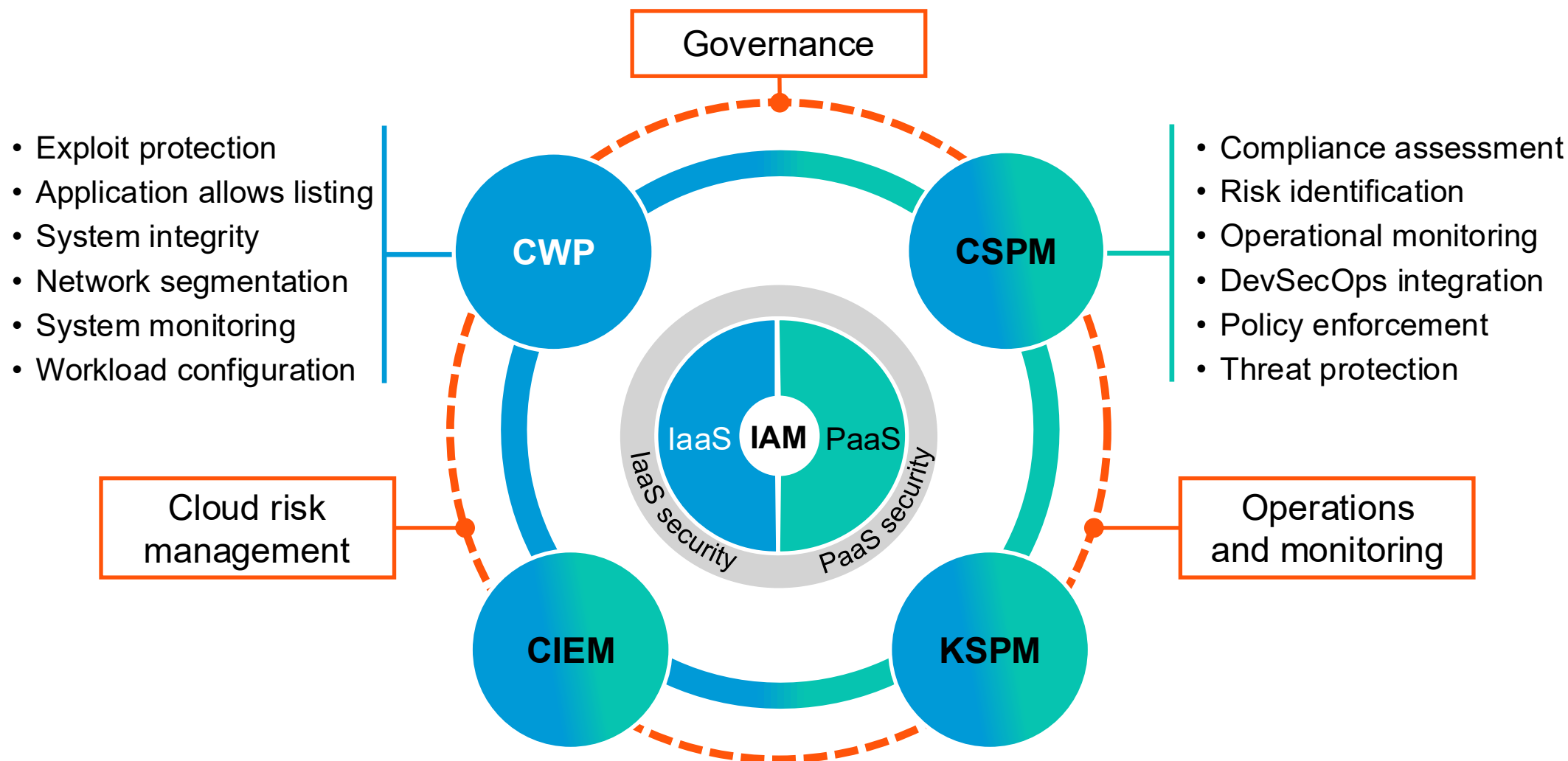
Configuration Includes Identity ...



Cloud Security Posture Management



Cloud-Native Application Protection Platform



CNAPP Capabilities

Required

- Runtime workload visibility
- CSPM and KSPM
- IaC scanning
- CIEM
- Container registry scanning
- Software composition analysis

Recommended

- Real-time workload visibility
- API discovery in dev and runtime
- Unstructured data risk
- Workload and cloud detection and response
- Drift detection
- Support for less common cloud providers
- Attack path analysis

Optional

- Serverless function support
- Application layer monitoring
- On-premises support
- Less common container orchestration support
- PaC support
- Structured data scanning
- AST technologies

AST = application security testing



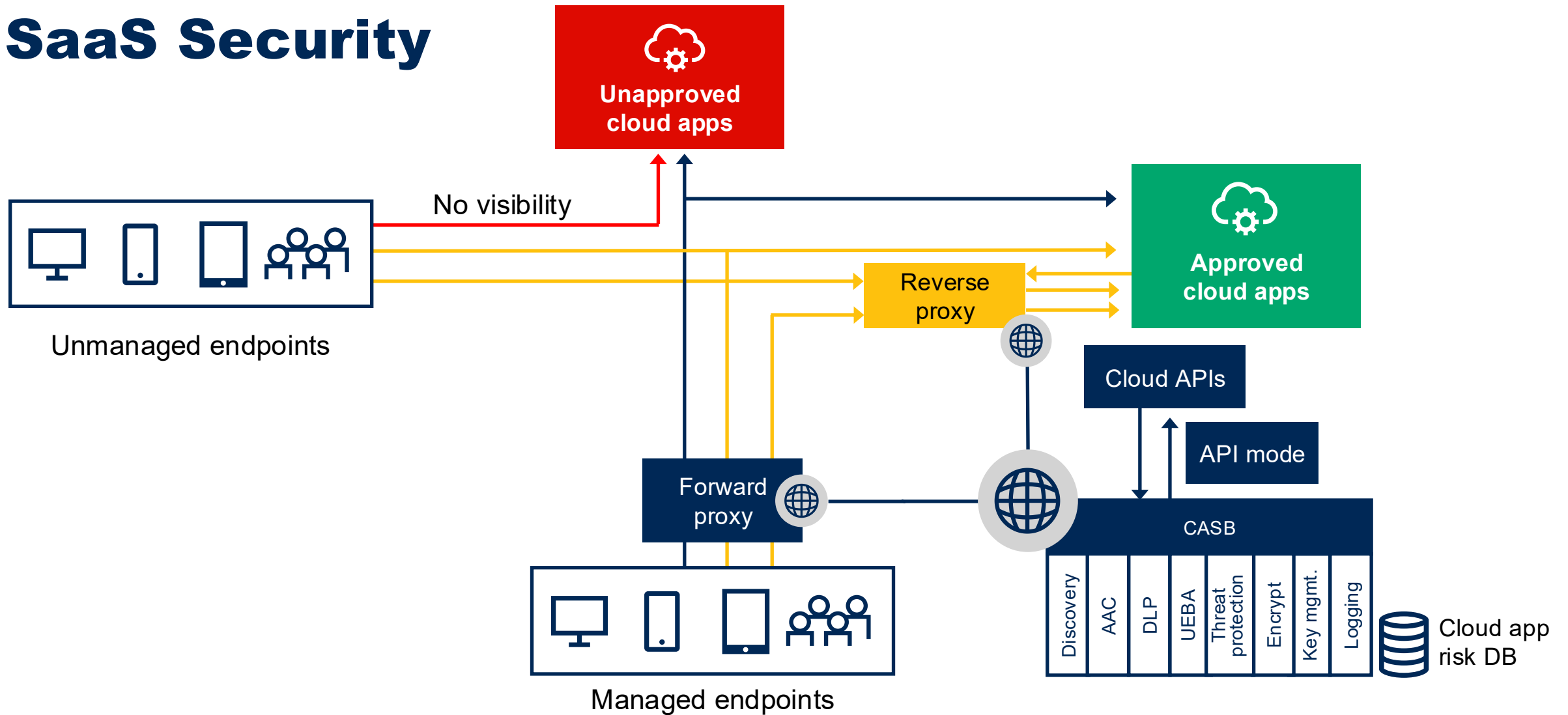
**In cloud,
network security controls
aren't enough.**

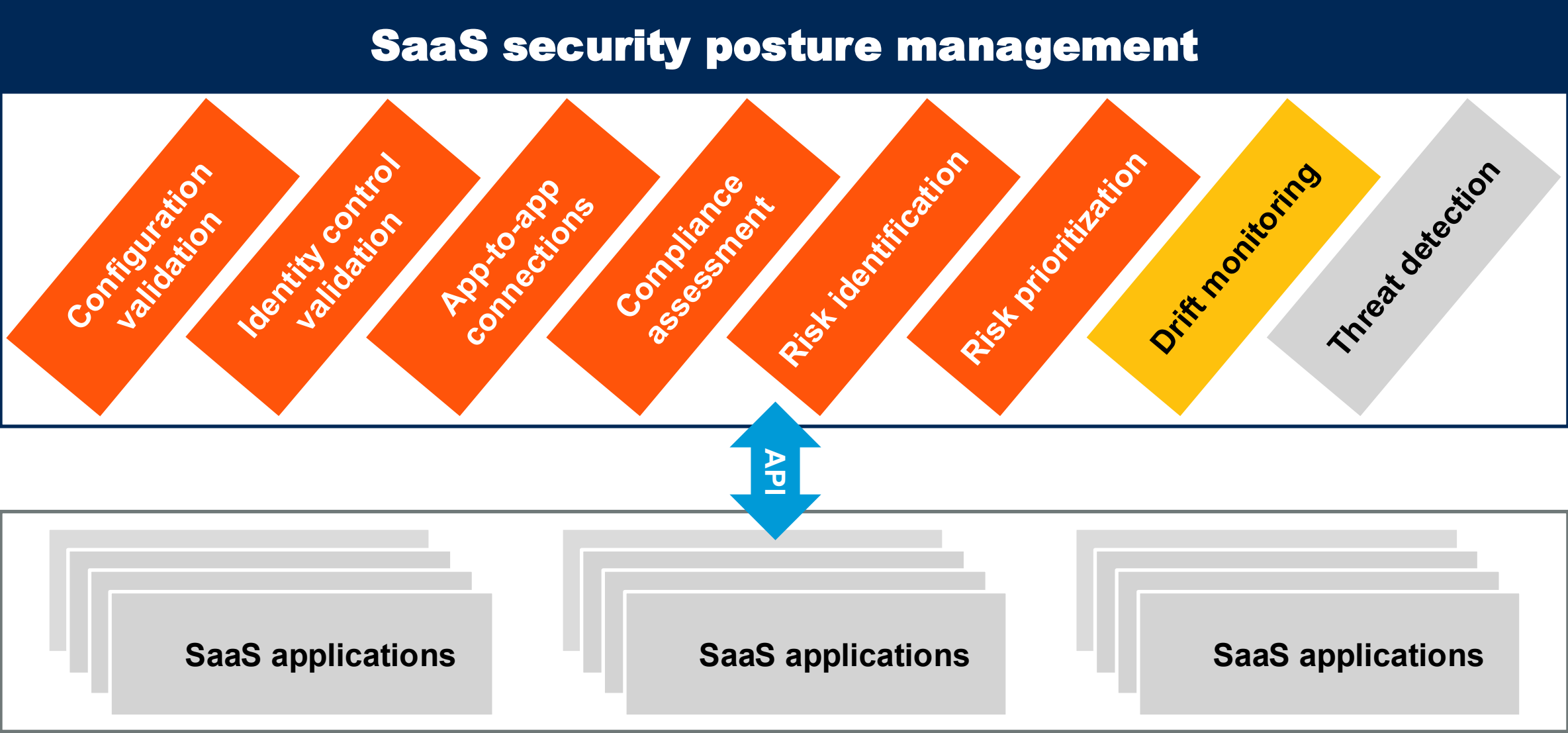


SaaS Controls

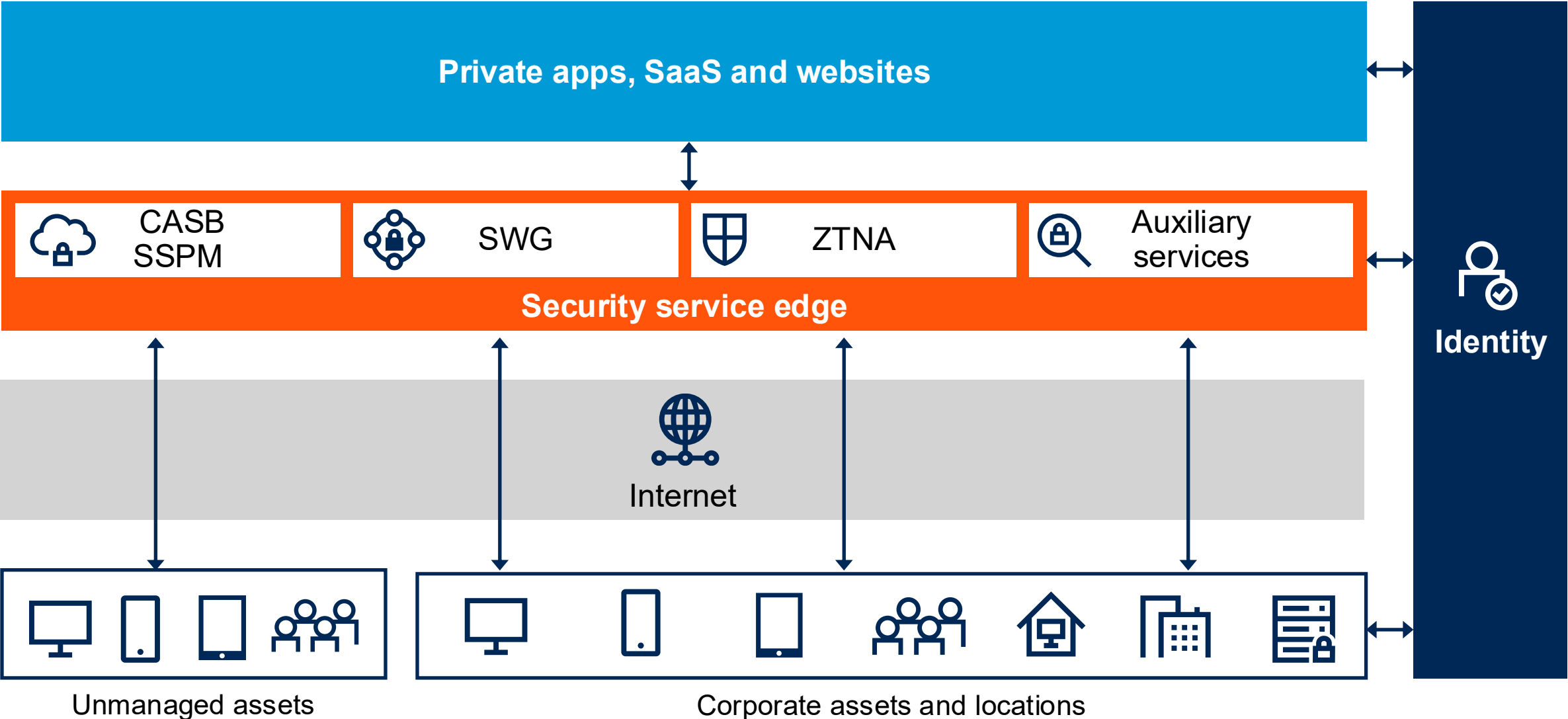
**Typical organizations have
over 200 SaaS applications —
all outside the perimeter.**

SSE for SaaS Security





Security Service Edge





Different cloud models need different controls and different approaches — and different to on-premises.

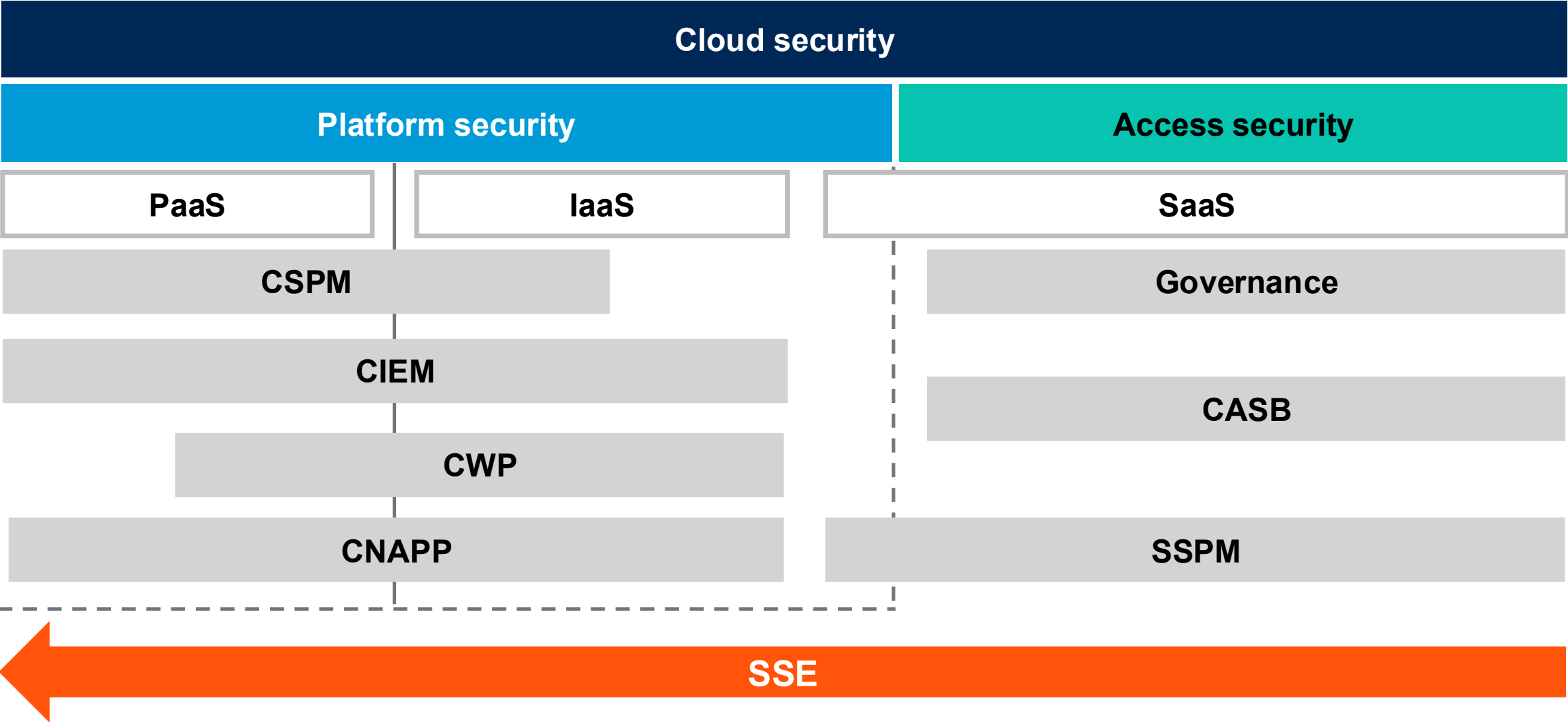
Where Do I Start?

Cloud security

Platform security

SSE

Where Do I Start?



Recommendations

- ④ Focus on how you operated the cloud, not the cloud provider's security.
 - Plan SaaS and IaaS controls differently (and differently to on-premises).
 - Prioritize automated posture and identity visibility in your most critical clouds.
- ④ Start with CSPM for a hyperscaler and CASB in SaaS.
- ④ Consolidate tools into a set of platforms, not a single vendor.
 - Evaluate SSE for SSPM.
 - Evaluate CNAPP for CSPM and more.
- ④ Be open to composing platforms, especially CNAPP and CWP.

Recommended Gartner Research

To learn more about access to Gartner research, expert analyst insight, and peer communities, contact your Gartner representative or click on “Become A Client” on gartner.com to speak with one of our specialists.

🔍 **[Quick Answer: How to Choose the Right Security Posture Management Tools for Your Clouds](#)**

Charlie Winckless, Manuel Acosta and Dale Koeppen

🔍 **[Solution Path for Security in the Public Cloud](#)**

Richard Bartley, Fred Sotolongo and William Dupre

🔍 **[How to Protect Your Clouds With CSPM, CWPP, CNAPP and CASB](#)**

Fred Sotolongo and Richard Bartley